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AI for Bharat Hackathon

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SYNDICATE

Team Name : SYNDICATE

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Problem Statement : TrustGraph is an agentic AI infrastructure that provides informal workers with a secure, verifiable, and portable digital identity. It establishes a "Unified Trust Layer" to bridge the gap between simple information access and true financial inclusion.

Brief about the Idea:

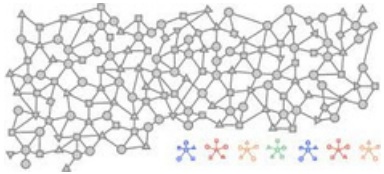
The TrustGraph Engine is an agentic AI infrastructure designed to solve the "Trust Deficit" for India's 490 million informal workers. It creates a Unified Trust Layer by converting intangible "word-of-mouth" reputations into cryptographically secure, verifiable digital identities.

By layering directly onto India's Digital Public Infrastructure (DPI)—Aadhaar, UPI, and DigiLocker—the engine allows workers to own their professional history through a mobile digital wallet. This portable identity transforms years of unrecorded labor into bankable digital assets, finally unlocking access to formal micro-credit, social security, and safe, milestone-based employment.

- From Invisible to Bankable: It turns social proof (testimonials, work history) into a cryptographic "Trust Card" recognized by financial institutions.
- DPI-Native: It doesn't replace existing systems; it enhances them by adding a reputation layer to the India Stack.
- Empowerment, Not Just Access: It gives the worker sovereign ownership of their data, moving them from being "beneficiaries" to active catalysts of a Viksit Bharat.

How different is it from any of the other existing ideas?

- **From Data Entry to Data Intelligence:** Traditional portals (like LinkedIn or standard job boards) rely on manual, text-heavy data entry. TrustGraph uses Graph Neural Networks (GNNs) to analyze "proxy data"—such as UPI transaction consistency and voice-verified testimonials—to build creditworthiness without needing physical collateral.
- **From Information Access to Economic Identity:** Most existing apps provide a list of jobs but no way to prove a worker's reliability. TrustGraph converts "word-of-mouth" social proof into Verifiable Credentials (VCs) that are cryptographically signed and bankable.
- **From High Barriers to Zero Friction**: Current DPI tools often require high digital literacy. TrustGraph integrates a Voice-First Vernacular AI (via Bhashini), allowing workers to interact naturally in their own dialect, making the technology invisible and the impact immediate.



BHASHINI

How will it be able to solve the problem?

- **Formal Digital Footprint:** Converts undocumented work and skills into tamper-proof Verifiable Credentials (VCs), creating an "economically visible" history for the 90% of workers currently without formal proof.
- **Reputation-Based Credit:** Replaces traditional collateral with "Reputation Capital." It uses GNNs to analyze social proof and UPI patterns, generating a Resilience Score that banks can trust for lending.
- **Guaranteed Wage Security:** Employs Agentic Smart Contracts for instant, milestone-based payments. Verification via geotagged photos triggers auto-disbursal, eliminating middleman exploitation and wage delays.
- **Vernacular Accessibility:** Removes digital literacy barriers through Bhashini-integrated Voice AI. Workers build their "Trust Graph" simply by speaking in their native dialect, making formalization effortless.

UPI UNIFIED PAYMENTS INTERFACE | BHASHINI | DigiLocker

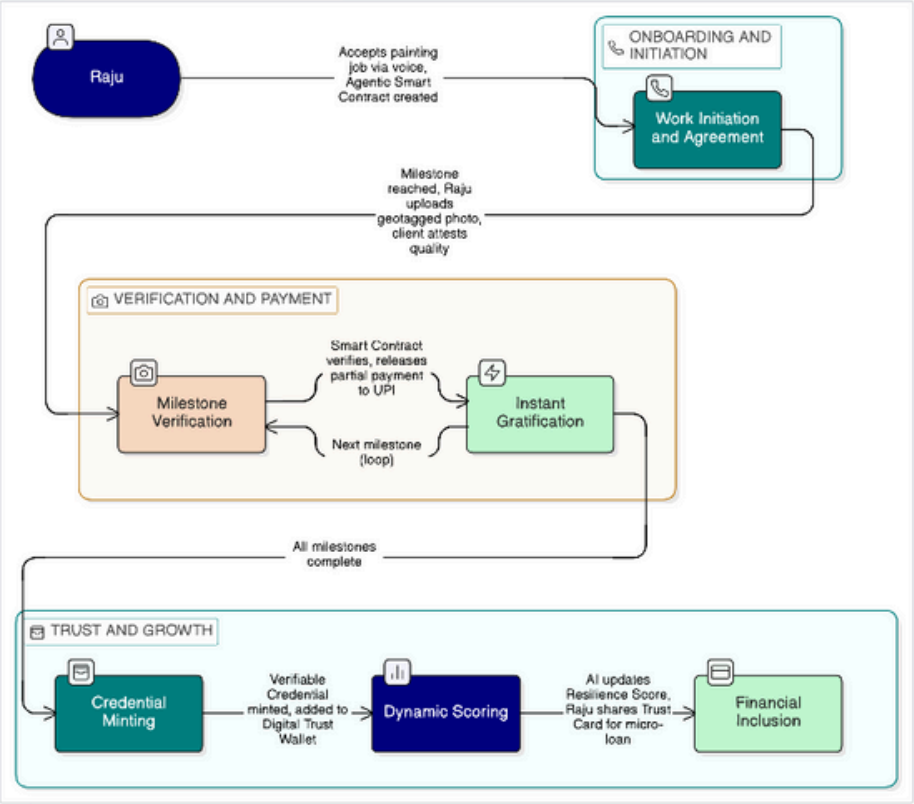
USP of the proposed solution?

- **Zero-Barrier Accessibility:** Unlike text-heavy portals, our Voice-First Vernacular AI (powered by Bhashini) makes the entire system accessible to the 490 million informal workers through natural speech in their native dialects, removing all literacy and technical barriers.
- **Programmable Trust:** By integrating Milestone-Based Smart Contracts, we don't just record work; we secure it. Payments are automatically triggered by verifiable proof (geotags/photos), ensuring 100% wage security and eliminating middleman exploitation.
- **Alternative Credit Ecosystem:** We replace traditional collateral with "Reputation Capital." Our Graph Neural Network (GNN) models analyze transaction consistency and verified testimonials to create a Resilience Score, making the unorganized workforce instantly "bankable" for formal micro-credit.

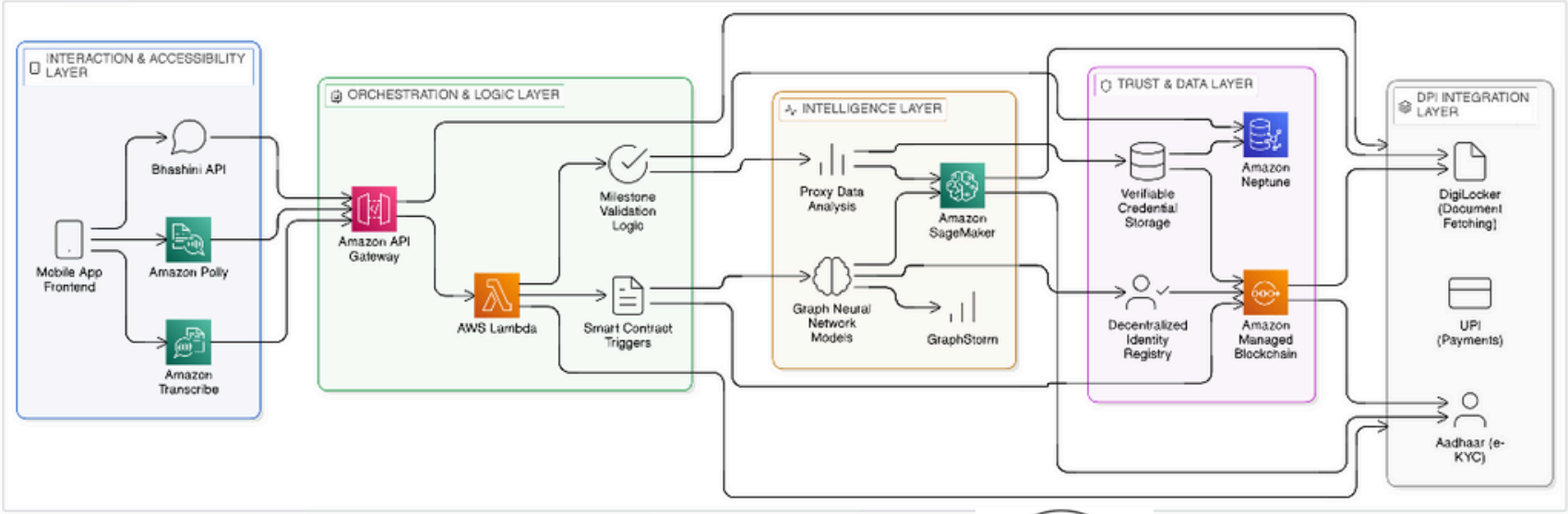
List of Features

- Voice-First Interface: 22+ Bhartiya language support via Bhashini, allowing workers to "talk" to the system to log work, removing all literacy barriers.
- Digital Trust Wallet: A secure, self-sovereign space where workers own their "Reputation Assets" (Verifiable Credentials) issued by employers.
- Predictive Resilience Score: A dynamic credit score powered by Graph Neural Networks (GNNs) that analyzes social proof and payment patterns instead of traditional collateral.
- Agentic Smart Contracts: Self-executing agreements that auto-disburse payments instantly upon milestone verification (geotags/photos).

Process flow diagram



Architecture diagram of the proposed solution:



Technologies used

1. The Brain (Intelligence & Accessibility)

- Amazon SageMaker: Deploys GNNs to analyze social patterns and generate credit resilience scores.
- Bhashini API: Enables ASR/TTS in 22+ Bhartiya languages, removing literacy barriers.
- Amazon Polly: Provides natural, context-aware audio guidance for intuitive interaction.

2. The Ledger (Trust Infrastructure)

- W3C Verifiable Credentials: Creates portable, cryptographically signed professional records.
- Amazon Managed Blockchain: An immutable ledger to host reputation identities and work milestones.
- AWS KMS: Manages the cryptographic keys for secure signing and data protection.

3. The Foundation (Architecture & Data)

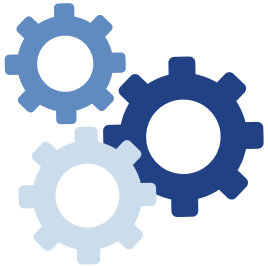
- Amazon Neptune: Graph database mapping intricate worker-employer-financial relationships.
- AWS Lambda: Serverless execution of Agentic AI and automated payment disbursals.
- Amazon S3: Secure, scalable storage for geotagged job evidence and digital wallets.

Estimated implementation cost

1. Prototyping (Phase 0):
   - Cost: Zero/Minimal.
   - Strategy: Leveraging AWS Free Tier and SageMaker Trial credits to develop and train initial Graph Neural Network (GNN) models.
2. API & Integration (Phase 1):
   - Cost: Optimized.
   - Strategy: Utilizing government sandbox environments for Bhashini and DPI (Aadhaar/UPI) integrations, keeping initial testing costs near zero.
3. Infrastructure & Scalability:
   - Cost: Pay-as-you-grow.
   - Strategy: A Serverless Architecture powered by AWS Lambda ensures operational costs only scale in direct proportion to the number of workers onboarded.
4. Development Timeline:
   - Full-Scale Functional Pilot: Estimated 6 months to launch high-impact use cases in initial pilot states.

Additional Hackathon Requirements

- Alignment: Directly addresses the Viksit Bharat @2047 vision and NITI Aayog's Digital ShramSetu mission roadmap.
- Centenary Impact: Focuses on increasing worker productivity from $5 to $49 per hour and elevating annual per capita income to $14,500.
- Social Security: Targets 100% universal social security coverage for the informal workforce by bridging the "Trust Deficit" with verifiable technology.



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Thank You